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Experimental and Numerical Investigation of Hydrogen Combustion in a Dual-Swirl Burner for Aero-Engine Applications

Green hydrogen produced by electrolysis offers a high potential for reducing CO_2 emissions and thus represents a promising approach for the decarbonization of aviation. However, propulsion systems based on direct hydrogen combustion require modified fuel injectors and combustion chambers to account for the particular combustion characteristics of hydrogen. Engineering those modifications requires the acquisition of experimental and numerical tools especially suited for this task and in the end validating them in a suitable environment. In this context, hydrogen combustion and its numerical simulation are presented with a dual-swirl burner in an optically accessible atmospheric combustor as an intermediate step. To ensure safe operation and to reduce the risk of flashback, fuel and air are injected nonpremixed. Good flame stability and mixing, which leads to potentially low NO_x values, is achieved by introducing a swirling motion into the flows. In this study, the combustor is operated under atmospheric pressure at a globally lean equivalence ratio. Measurements of OH radical chemiluminescence as well as infrared (IR) radiation as marker of the hot water vapor distribution have been carried out to identify the flame location and shape. The configuration is further analyzed by means of reacting large-eddy-simulations (LES). The comparison of the simulation results with the experimental reference data shows that the flame lift of height and global flame spread are correctly predicted by the simulation for both operating conditions. However, the combustion model does not precisely capture the flame stabilization mechanism, leading to a radial offset of the flame front.*
[DOI: 10.1115/1.4065925]

Keywords: hydrogen combustion, swirl stabilized flames, optical measurements, numerical modeling

1 Introduction

In the face of the climate crisis, greenhouse gas emissions must be reduced across all sectors. Reducing emissions from aviation is particularly challenging because solutions which add weight or volume directly reduce system efficiency. In this context, direct combustion of hydrogen is one of the promising concepts to decarbonize aircraft engine propulsion systems. The use of green hydrogen produced by electrolysis using renewable energy completely eliminates direct carbon dioxide emissions. Although hydrogen has a high gravimetric energy density, its volumetric energy density is low. This poses significant challenges for the technical infrastructure and, in particular, the storage of hydrogen on board of an aircraft [1]. Despite these unresolved challenges, hydrogen offers the potential to replace hydrocarbon fuels, particularly for short- and medium-range applications.

In addition to fuel handling challenges, fundamental combustion system adjustments are expected due to the unique properties of hydrogen. The combustion characteristics of hydrogen, such as stoichiometric flame temperature, flammability limits, and flame

Turbo Expo June 24–28, 2024. GT2024.
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Manuscript received June 20, 2024; final manuscript received July 3, 2024; published online August 16, 2024. Editor: Jerzy T. Sawicki.

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speed, differ significantly from those of commonly used hydrocarbon fuels [2]. The higher stoichiometric flame temperature of hydrogen may result in higher peak temperatures and therefore higher nitrogen oxide (NO_x) emissions and higher thermal loads, although the exit temperature would be similar for the same thermal output. Hydrogen's high reactivity, and therefore higher flame speed, results in compact flames and increases the risk of flashback. On the other hand, hydrogen has the potential to even reduce NO_x emissions compared to hydrocarbon fuels [3]. The very wide flammability range reduces the risk of flame extinction and allows safe operation at conditions far away from stoichiometry. Furthermore, the compact flames offer the potential to reduce the size of the combustion chamber and the residence time in the hot primary zone. A side effect of smaller combustion chambers is a reduced demand of cooling air due to less heat loaded surfaces.

Thus, the performance of future hydrogen propulsion systems is highly dependent on the specific design of the combustion system. Therefore, the exploration of a limited variety of approaches might be needed to arrive at the best solutions. Furthermore, a better understanding of the specific characteristics of hydrogen combustion and its impact on aero combustor design is needed. In this context, hydrogen combustion and its numerical simulation are investigated in this work with a dual-swirl burner in an optically accessible atmospheric combustor.

Several decades ago, the use of hydrogen in aircraft engines led to a series of investigations [4,5]. These early explorations were primarily motivated by the threatened depletion of fossil fuels. With the need for decarbonization of the aviation industry, this topic is more relevant now than ever. Despite these early works, the use of hydrogen in aircraft engines is still a relatively new area of research. In comparison, considerable work has been carried out on the use of hydrogen and fuel blends (e.g., with natural gas) in ground-based gas turbines [6]. The limiting factors for combustion of pure hydrogen in typically premixed dry low-NO_x combustors are flashback and thermoacoustic instabilities. Diffusion burners of industrial gas turbines have been operated with pure hydrogen, but typically require water injection to keep the NO_x emissions within limits. A widely investigated approach are so-called micromix combustors [7,8]. A multitude of small channels containing jet-in-crossflow hydrogen injectors are clustered to achieve distributed diffusion flames, reducing NO_x emissions by limiting residence time in high-temperature regions. While this concept has recently been brought into a commercial burner in the context of stationary gas turbines [9], the feasibility for aircraft engines and the thermoacoustic risks are currently under investigation [10,11].

Considering the various challenges of such new developments, adapting existing concepts seems most promising for a rapid transition to pure hydrogen combustion in aviation. Conventional aero-engine combustion systems rely on swirling flow for flame stabilization and separate injection of fuel and air to prevent flashback. While previous studies have operated several swirl burner configurations with hydrogen-enriched fuels in the context of stationary gas turbines [12,13], a large volume fraction of hydrogen is required to significantly alter the combustion characteristics. However, the available literature on swirl burners running on pure hydrogen is still very limited. A premixed configuration has been studied experimentally by Reichel et al. [14] and by means of numerical simulation [15,16]. To the best of the authors' knowledge, the HYLON configuration, which has been investigated both experimentally [17,18] and numerically [19], is the only test case in the open literature that combines swirl stabilized combustion of pure hydrogen with separate injection of fuel and air.

The configuration studied in this paper was developed in the context of the rich–quench–lean concept. The fuel injector was originally designed to simulate kerosene combustion with natural gas at atmospheric conditions by producing a gas film with mixing properties similar to a kerosene film. For this study, it was operated with pure hydrogen at lean equivalence ratios.

The investigated test case operates in a partially premixed mode, where both premixed and nonpremixed type regimes occur. The simulation of such configurations is particularly challenging for numerical tools. Within the framework of tabulated chemistry combustion models, hybrid approaches based on both premixed and nonpremixed flamelets have been proposed [20,21]. For the HYLON burner simulations, Aniello et al. [19] use a finite rate chemistry approach with the dynamic thickened flame (DTF) model and disable flame thickening in nonpremixed regions identified by the Takeno flame index [22]. In this paper, the introduced configuration is further analyzed using reacting large-eddy-simulations (LES) with a standard flamelet-generated manifolds (FGM) combustion model based on premixed flamelets in combination with the DTF model. This modeling approach has been successfully applied to the prediction of a swirling natural gas flame [23]. It should be noted that the nonpremixed regime is not explicitly considered by either the FGM model or the DTF model, which was developed for premixed flames only. However, this work aims to test the suitability of this standard combustion modeling approach for predicting the combustion mode and flame characteristics in a multiregime hydrogen combustion system.

The paper is organized as follows: The experimental setup is presented in Sec. 2, including the combustor test rig, the operating conditions, and the optical measurement techniques. The numerical modeling approach is described in Sec. 3. Numerical and experimental results are presented and analyzed in Sec. 4, followed by concluding remarks in Sec. 5.

2 Experimental Setup

This section presents the experimental setup. The burner test rig and the operating conditions are described first, followed by a summary of the measurement techniques used.

2.1 Burner Test Rig. The experimental campaign was conducted in the atmospheric primary zone [24,25] facility of the DLR Institute of Propulsion Technology in Cologne, Germany. The facility has recently been upgraded for the study of hydrogen combustion and can be continuously supplied with up to 1.2 g/s of hydrogen. In this test bed, a dual-swirl burner in an optically accessible atmospheric combustion chamber is investigated. A detailed view of the burner configuration is presented in Fig. 1. The combustion chamber cross section is hexagonal with interchangeable wall segments. Each wall segment is 55 mm wide and 85 mm high. The injector was originally operated with natural gas and is used in this study for the combustion of pure hydrogen.

A sketch of the fuel and air flow paths is shown in Fig. 1(*a*). The preheated primary air enters the test section at the bottom. Most of the primary air then passes through a corotating dual-swirler configuration. A smaller amount which is by-passing the burner through small metering holes is used for convective cooling of the heat shield and providing a thin air film on the combustor windows. The flow path along the inner heat shield wall is equipped with small pin fins to enhance the heat transfer from heat shield to cooling air.

A rotating velocity component is imposed on the burner air by the corotating inner and outer radial swirlers. The outer swirler diameter is $d_o = 23.7$ mm, and the inner swirler diameter is $d_i = 13.7$ mm. The air swirlers and the burner exit diffusor can be exchanged due to the modular design of the burner. A high swirl for the outer and a low swirl for the inner air stream combined with a cylindrical diffusor were chosen to keep the flame compact. For this first hydrogen campaign, a design that limits the temperature load on the walls seemed advisable. The measured effective area of the installed burner is 251.8 mm^2.

Hydrogen at ambient temperature is supplied to the burner via a central supply line. It then passes through a series of eight distributed channels to a circumferential fuel plenum. From here, the hydrogen is directed into the fuel swirler through 16 radial holes of 1 mm diameter to ensure circumferential homogeneity. Finally, the hydrogen is axially injected into the circumferential shear layer between the inner and outer burner air with a corotating velocity

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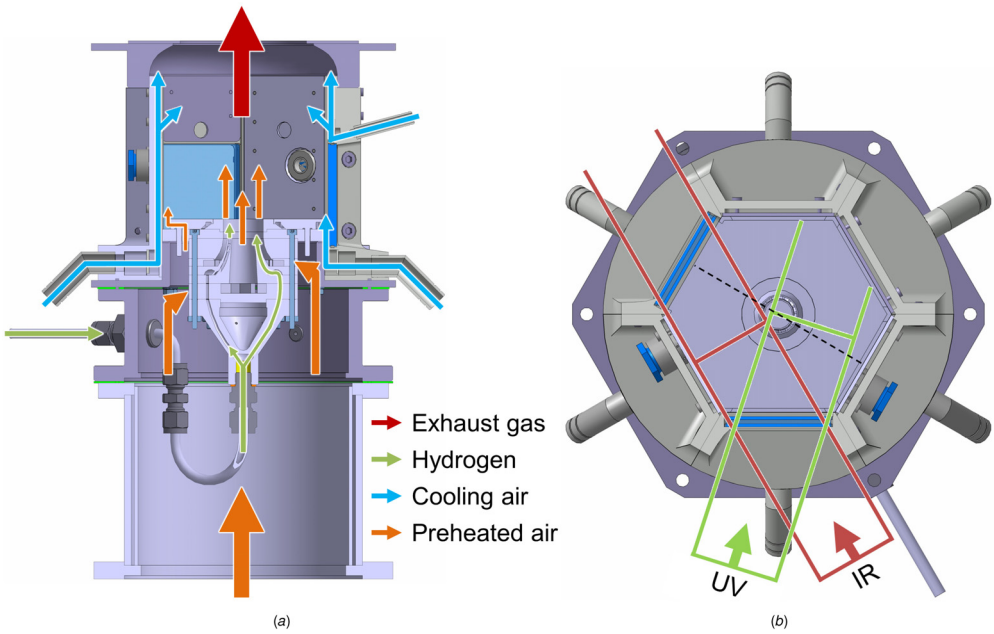

Fig. 1 Burner test rig: (*a*) longitudinal section with illustration of the different flow paths and (*b*) horizontal cut with illustration of the respective camera angles (the dashed line shows the analysis plane used in the results section for OH*)

component. The inner and outer diameters of the annular fuel gap are $d_i = 14.9$ mm and $d_o = 17.9$ mm, respectively.

Additional supply lines provide ambient air to cool the walls and windows of the hexagonal combustion chamber. For the four metal wall segments, the air is supplied surrounding the burner base plate and provides convective cooling from outside the combustion chamber. It is partially injected into the secondary zone through dilution holes of 9 mm in diameter and partially enters the combustion chamber just upstream of the outlet. The two combustion windows for measurement access are supplied with cooling air through two separate channels: An upstream supply line injects air at the bottom of the combustion chamber where it mixes with the heat shield cooling air to form a cooling film along the windows within the combustion chamber. The second channel provides additional air to cool the downstream metal wall segment. This air is injected into the combustion chamber through the dilution holes as well as downstream near the outlet.

A converging section is attached to the combustion chamber outlet to prevent any backflow at the outlet.

2.2 Operating Conditions. The burner is operated at atmospheric pressure with a lean equivalence ratio. Two operating conditions with different fuel mass flow rates are considered in this study, as summarized in Table 1. Keeping all other parameters identical between the two operating conditions results in different heat outputs and equivalence ratios. The equivalence ratios Φ_{PZ} presented in Table 1 are for the primary zone and consider only the burner air.

The Reynolds number for both operating conditions is similar due to the same air mass flow. The Reynolds number based on the air mass flow and the outer diameter of the injector is about 55,000.

The hydrogen and the downstream cooling air are injected at an ambient temperature of 291.65 K, while the primary air is preheated to 376.2 K. The primary air mass flow rate is 22.17 g/s, and the combined cooling air mass flow rate for the four metal wall segments is 10.08 g/s. The cooling air mass flow for the two windows is 3.96 g/s for the cooling film and 2.28 g/s for the convective cooling of the downstream metal section.

Table 1 Operating conditions

	$\dot{m}_{H_2}$ (g/s)	P_{th} (kW)	Φ_{PZ}	Δp (%)
Flame A	0.498	59.8	0.895	2.991
Flame B	0.301	36.1	0.543	2.932

2.3 Measurement Techniques. The two window segments of the atmospheric primary zone provide optical access to the primary zone. OH* radical chemiluminescence as well as infrared (IR) radiation as a marker of the hot water vapor distribution have been measured using an ultraviolet (UV) and IR camera, respectively. The camera positions and orientations are sketched in Fig. 1(*b*), with the cameras generally oriented perpendicular to the flame. The UV camera (PCO Dicam Pro, 1280 × 1024 pixel with 2 × 2 binning, 100 mm UV lens, f/2.8), which records OH* chemiluminescence in the spectrum from 305 to 325 nm, is oriented parallel to the heat shield and angled slightly inward to capture the burner centerline. The infrared camera (Xenics Xeva vSWIR, 320 × 256 pixel InGaAs detector with spectral range 500–1700 nm, f/2) records the spectrum through a spectral bandpass filter with a central wavelength of 1425 nm and a full width at half maximum of 190 nm, which covers relevant water vapor radiation bands. The optical path with windows on both combustor sides is chosen to minimize thermal background radiation from hot combustor walls. With this setup, approximately half the width of the combustion chamber is covered by both measurements. Both cameras record 400 individual line-of-sight images, which are normalized by exposure time and averaged into a single mean image for the presented analysis. The images are additionally deconvoluted using the PYABEL package and the basex method [26] to obtain center cut views. Rotational symmetry of the flame, a prerequisite for the application of deconvolution methods, is assumed to be sufficiently given due to the hexagonal cross section of the combustion chamber.

The optical line-of-sight measurements are nonintrusive and relatively simple. Spontaneous OH* emissions from flames, which are caused by excited OH* radicals returning back to a lower energy state, are widely used as a marker of the flame shape and position [27,28]. The radiation of water vapor, as the main product of hydrogen combustion, is expected to increase along the flame. However, as shown by Rivière and Soufiani [29], the emitted radiation not only depends on the water vapor concentration but also

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temperature and pressure. Hence, the recorded intensity in the experiment increases with concentration and temperature. The individual effects on the emissivity are very different, and the images recorded during the experiment must be interpreted accordingly.

3 Simulation Setup

The burner test rig described in Sec. 2.1 is further analyzed by means of numerical simulations. In this section, the numerical model is summarized, followed by a description of the computational domain and boundary conditions.

3.1 Numerical Model. Large-eddy-simulations are performed with DLR's in-house computational fluid dynamics (CFD) solver TRACE. The simulations are based on the FGM combustion model [30]. The chemistry tabulation is performed in a preprocessing step based on adiabatic one-dimensional premixed freely propagating flames using CANTERA [31]. The hydrogen branch of the San Diego chemical mechanism [32] has been used, which consists of 9 species and 21 reactions. For direct comparison with the OH* radical chemiluminescence measurements, the mechanism has been extended with the OH* submechanism of Kathrotia et al. [33]. The coupling strategy of the isobaric FGM tabulation with the fully compressible flow solver is presented in Ref. [23].

The DTF model [34] is used to resolve the flame front on the LES mesh. The modified [35] power-law flame wrinkling model of Charlette et al. [36] is used as efficiency function. The intensity of the subgrid velocity fluctuations is computed as proposed in Ref. [37]. A flame sensor based on the normalized reaction progress variable is used to limit the thickening to the flame region [37]. The number of grid points to resolve the flame is set to $n_p = 5$.

A unity Lewis number assumption has been made to simplify the numerical model. Due to the high Reynolds number, turbulent transport is expected to dominate over molecular diffusion, limiting the influence of the Lewis number assumption on the simulation results.

The wall-adapting local eddy-viscosity model [38] is used as the turbulent closure for the subgrid Reynolds stresses. The numerical discretization is based on a central kinetic energy and entropy preserving scheme using the split form of Shima et al. [39] for the convective fluxes in combination with Burg's unstructured monotonic upstream-centered scheme for conservation laws reconstruction [40] and the limiter function of Venkatakrishnan [41]. Upwind fluxes are used for the convection of reactive scalars and the internal energy to limit total variation. The Jameson–Schmidt–Turkel matrix-valued artificial dissipation model [42,43] is applied to stabilize the solution. Further details on the numerical discretization can be found in Ref. [44]. The least-squares fully augmented approach described in Ref. [45] is used to calculate cell centered gradients, while the derivatives for the viscous fluxes are computed based on the face neighboring cell gradients with the face-tangent augmentation outlined in Ref. [46]. A third-order accurate explicit Runge–Kutta method is used for time integration.

3.2 Simulation Setup. The geometric complexity of the burner test rig presented in Sec. 2.1 has been slightly reduced to perform the numerical simulations. The derived computational domain with the different boundary patches is presented in Fig. 2(*a*). While the air and fuel plenum are included in the computational domain, the secondary flow paths of the cooling air are not explicitly resolved by the simulation, but are replaced by equivalent boundary conditions. Specifically, the boundary conditions for the simulation are defined as follows: The air mass flow through the burner (injected at the main air inlet in Fig. 2(*a*)) is calculated from the measured primary air mass flow and the burner effective area determined in a preceding measurement. The remaining primary air is used for convective cooling of the heat shield in the measurements. The flow path along the inner heat shield walls is not included in the computational domain but the respective mass flow is injected through horizontal boundary patches at the side walls of the combustor (heat shield and window cooling air inlet patches in Fig. 2(*a*)). At the window segments, additional cooling air is injected upstream of the glass windows, and the additional mass flow is added to the respective heat shield air boundary patches. The cooling air that is used to cool the outer chamber walls is introduced through additional boundary patches (dilution air inlet and downstream cooling air inlet patches in Fig. 2(*a*)). The mass flow split between the dilution ports and the cooling inlets at the combustion chamber exit is approximated based on their area ratio. In general, downstream air injection has little effect on the primary zone. The combined mass flows for all four metal and two window segments have been recorded during the experiments. The mass flow is evenly distributed among the different wall segments and prescribed as boundary conditions in the simulations. The direction vector at the dilution ports is assumed to be 45 deg with respect to the horizontal plane and prescribed at the boundary accordingly.

The mass flow rates and temperatures prescribed at the different boundaries are summarized in Table 2. Atmospheric static pressure is prescribed at the outlet. While mass flow rate boundary conditions are used for cooling air and fuel inlet, nonreflecting boundary conditions [23,47] are used at the air inlet and the outlet. All solid walls are modeled as adiabatic boundaries in combination with the wall functions of Reichardt [48].

The computational domain is discretized using an unstructured tetrahedral mesh with about 19.1×10^6 cells. The mesh is refined in the swirler, fuel supply, and flame region as shown in Fig. 2(*b*). The cell size in this region based on the cells' edge length is 0.5 mm.

A constant time-step of 1.5×10^{-8} s is used. To ensure a converged solution, the simulations were run for about 30 ms before the statistics were collected, corresponding to four flow times of the combustion chamber. Statistics were then collected for another 30 ms.

4 Results

In this section, the ability of the numerical model to predict the flame position and shape is first evaluated by comparison with the measurements. Then, the two operating conditions are further analyzed based on the simulation results. Finally, the flame index is used to determine the different flame regimes predicted by the simulations.

4.1 Comparison of Experiments and Simulations. The distributions of OH* radicals and hot water vapor radiation in the flame are available from the experiments. These measurements are used below to validate the numerical simulations.

4.1.1 OH Radical Chemiluminescence.* The OH* radical distribution can be used as a marker of the flame. It is therefore often compared with the heat release rate (HRR), which is easily obtained from CFD simulations. In this work, an OH* submechanism was additionally used to allow a direct comparison with the experiments. The mean distribution of normalized HRR and OH* obtained from the simulation is shown in Fig. 3 for flame A. To avoid any asymmetry influence, the simulation results are mirrored at the burner axis. The distributions for the two variables are almost identical, which indicates that both of them can be used as a flame marker under the current conditions.

The primary zone of flames A and B is shown in Fig. 4. The mean normalized OH* mass fractions from the simulation results are compared to the mean normalized OH* distribution from measurements to analyze the flame shape and position predicted by the simulation. In the top figures, the Abel deconvoluted OH* are compared with the OH* mass fractions extracted on the analysis plane shown in Fig. 1(*b*). To assess the influence of the Abel deconvolution, the unprocessed measurement results are compared to line-of-sight integrated simulation results in the bottom figures. Additionally, corresponding profiles at different axial heights above the burner exit plane are shown in Fig. 5 to allow for a more detailed

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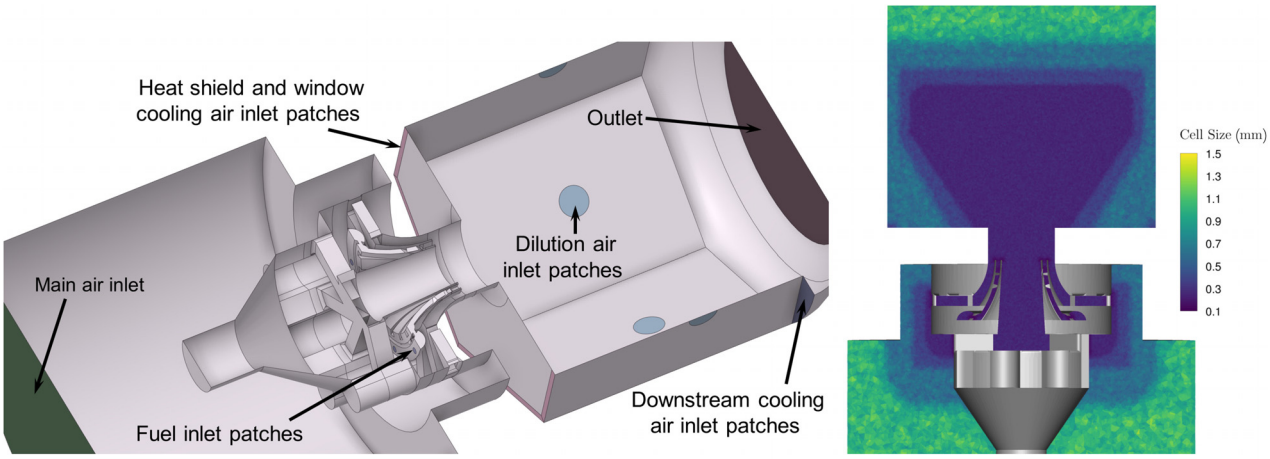

(*a*) (*b*)

Fig. 2 Computational domain: (*a*) longitudinal section of the computational domain with highlighted boundary condition locations and (*b*) cell sizes of the computational mesh

Table 2 CFD boundary conditions

Boundary	$\dot{m}$ (g/s)	T (K)
Air inlet	20.87	376.2
Fuel inlet	0.498/0.301	291.6
	Metal segments	
Heat shield air	0.217	376.2
Dilution air	1.034	291.6
Exit air	1.486	291.6
	Window segments	
Heat shield air	2.198	320.4
Dilution air	0.467	291.6
Exit air	0.671	291.6

comparison. In addition to the normalized OH* mass fractions, the simulated normalized HRR profiles are also included.

A lifted flame is observed for both operating conditions and correctly predicted by the simulations. In agreement with the experiments, the lift off height is slightly larger for the richer flame A. The global flame spread is reasonably well predicted by the simulations for both operating conditions.

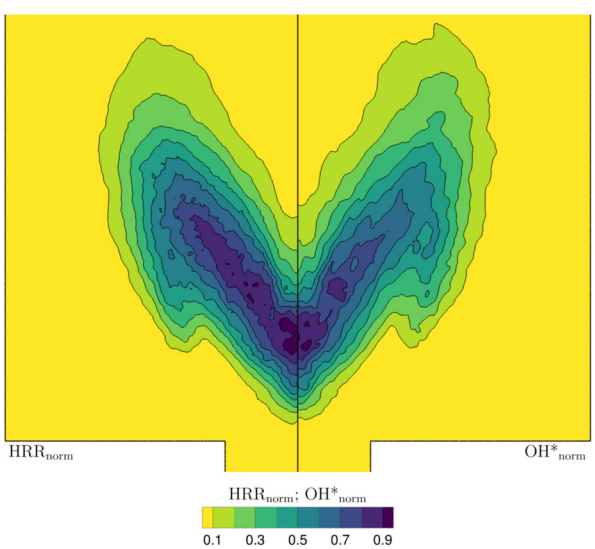

Fig. 3 Comparison of mean normalized heat release rate (left) and OH* (right) from the CFD simulation for flame A

Two reaction fronts can be identified for flame A: an inner front at the height of the burner exit plane and a lateral flame front further downstream. Both flame fronts are also predicted by the simulation, although the simulation does not predict an OH* maximum in the lateral flame front. The most significant difference between simulation and experiment is the different radial position of the inner flame front. While the flame front in the experiments is radially aligned with the inner wall of the fuel injector, the maximum OH* is predicted on the central burner axis in the simulation, indicating the flame stabilization at this location. In contrast, the experiments show no significant values on the centerline.

In agreement with the experimental data, the simulation of flame B stabilizes further upstream, and the flame front is located slightly inside the diffusor. The lateral flame front observed for flame A almost entirely vanished for flame B. While it is also significantly reduced in the simulation, it is still more pronounced compared to the measurements. Similar to flame A, the radial position of the flame front differs between simulation and experiments. As already observed for flame A, the flame is radially aligned with the inner wall of the fuel injector in the experiments, while it anchors on the central burner axis in the simulation.

The differences in the flame location between simulation and experiments indicate that the exact flame stabilization is difficult to predict with the combustion model used in this work. The flame front in the experiments is radially aligned with the mixing layer between inner air swirler and the fuel supply, which indicates a diffusion flame regime, that has not been captured by the simulation.

4.1.2 Hot Water Vapor Distribution. Infrared radiation as a marker of the hot water vapor distribution has been recorded using an IR camera. Since no corresponding quantity is directly available from the LES, the statistical band model parameters described in Ref. [29] are used to compute a lookup table for water vapor radiation in the spectral range of the filter, depending on temperature and water vapor mole fraction. The radiation intensity is then postprocessed from the simulation results using the lookup table. The normalized radiation values of simulation and measurement are compared in Fig. 6. As shown in Fig. 1(*b*), the water vapor radiation is measured and analyzed on a different plane compared to the OH* radical chemiluminescence.

For both operating conditions, the maximum hot water vapor radiation is observed on the centerline. In general, hot water vapor radiation occurs in regions where high H_2O concentrations coincide with high temperatures and can therefore be used as a marker of the postflame region. In this respect, the hot water vapor distributions confirm the observations from the OH* radical chemiluminescence presented in Sec. 4.1.1. While the general agreement between

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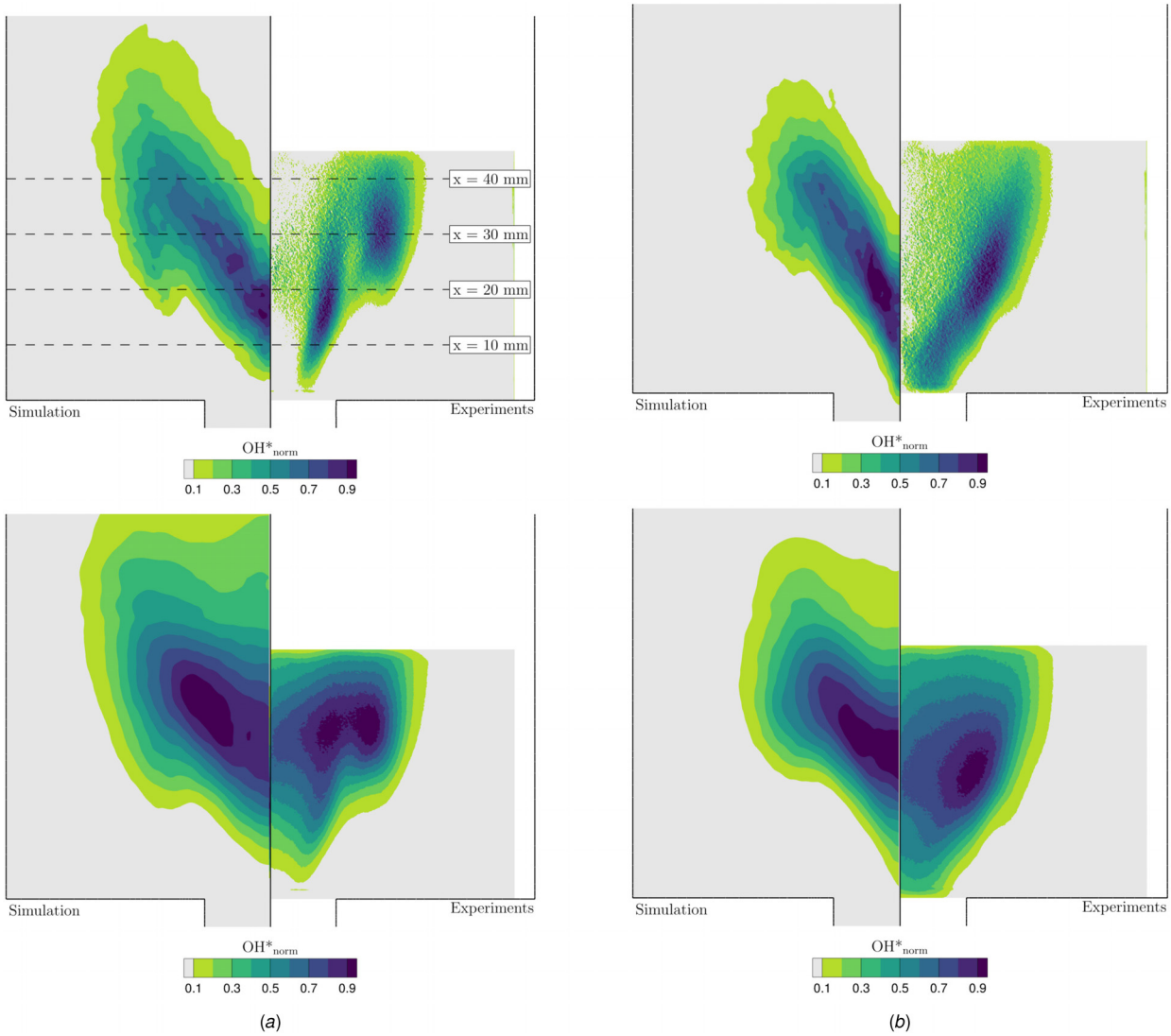

Fig. 4 Comparison of mean normalized OH* from simulations (left) and measurements (right): (*a*) flame A and (*b*) flame B. Top: Abel deconvoluted OH* measurement versus OH* on simulation plane, bottom: unprocessed OH* measurement versus line-of-sight integrated OH* from simulation. The dashed lines in the top left figure indicate the positions for the profiles in Fig. 5.

simulations and experiments is fairly good at downstream locations, differences can be found close to the burner exit plane due to the different radial position of the flame front. Since the flame is radially aligned with the fuel injector in the experiments, hot water vapor radiation occurs on the central axis already at the burner exit plane. In contrast, the simulations predict the flame front at the centerline, and consequently the hot water vapor radiation as an indicator of the postflame region occurs further downstream.

4.1.3 Pressure Loss. The pressure loss over the burner has been measured in the experiments using a pressure sensor in the plenum and in the combustion chamber at a wall downstream of the dilution ports. The pressure loss represents the relative differential pressure between these two measurement locations. In Table 3, the pressure losses from the mean LES fields are compared to the experimental data. The simulation overestimates the pressure losses for both operating conditions by 15.9%. Capturing the pressure loss with numerical simulations is generally nontrivial and would require significant mesh refinement in the swirler. However, Aniello et al. [19] observed a similar overestimation for a hydrogen swirl burner, even with a significantly finer mesh. Additional to the uncertainties associated with the numerical modeling itself, the pressure loss is directly linked to the air mass flow at the inlet boundary condition. This mass flow has been derived based on the effective area, which has been measured separately under cold conditions. Deviations of the selective laser melting manufactured burner and the computer aided design model used for the simulation represent an additional uncertainty. However, the small difference between the two operating conditions (the pressure losses between the two flames differ by about 2%) has been correctly reproduced by the simulation. Considering the good prediction of the flame lift of height and flame spread, the deviations in pressure loss are considered tolerable.

4.2 Comparison of Operating Conditions. In this section, the simulation results are further analyzed by comparison of the two operating conditions. Different mean fields are presented in Fig. 7 using the same plane as for the OH* comparison in Sec. 4.1.1. In each figure, flame A is on the left, while flame B is on the right-hand side. The plane has been mirrored at the burner axis for flame A to avoid any asymmetry influence.

In Fig. 7(*a*), the normalized heat release rate is presented. Zero axial velocity isocontours ($U_x = 0$) indicate the recirculation zones. Isocontours of mixture fraction mark stoichiometry (Φ_{st}) as well as lean (Φ_{lean}) and rich (Φ_{rich}) flammability limits according to the chemistry tabulation. Both flames are lifted above the burner due to an aerodynamic stabilization. The flame lift off results in an axial distance between the hydrogen injector lip and the flame front and

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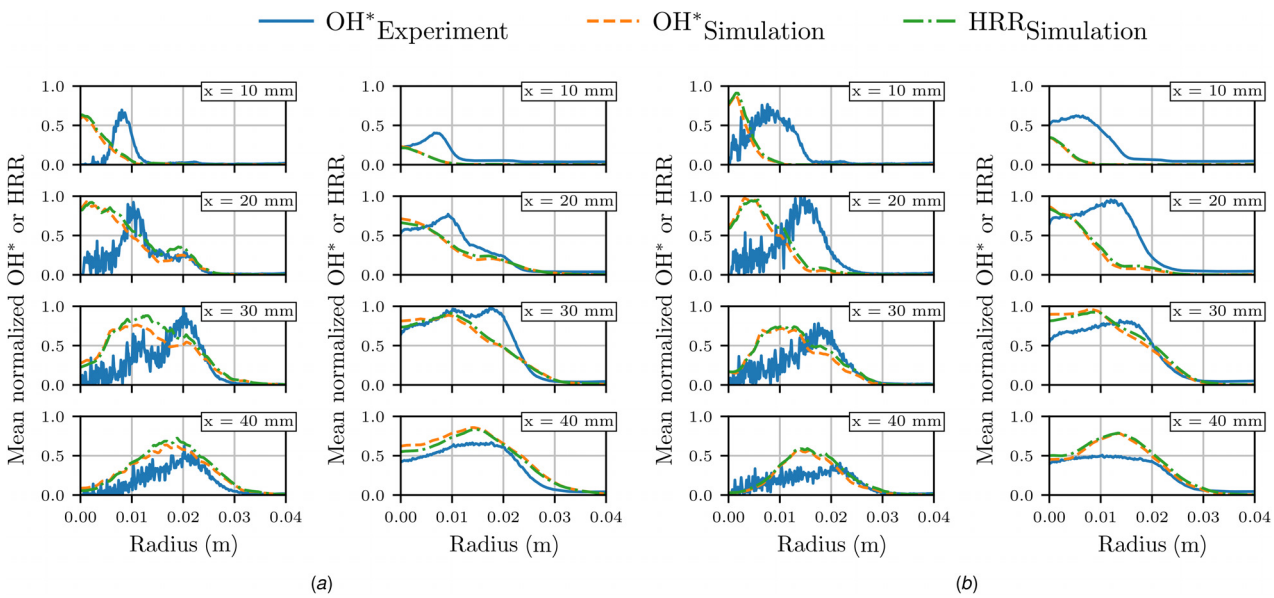

Fig. 5 Profiles of mean normalized OH* and HRR from simulations and mean normalized OH* from measurements: (*a*) flame A and (*b*) flame B. Left columns: Abel deconvoluted OH* measurement versus OH* on simulation plane, right columns: unprocessed OH* measurement versus line-of-sight integrated OH* from simulation.

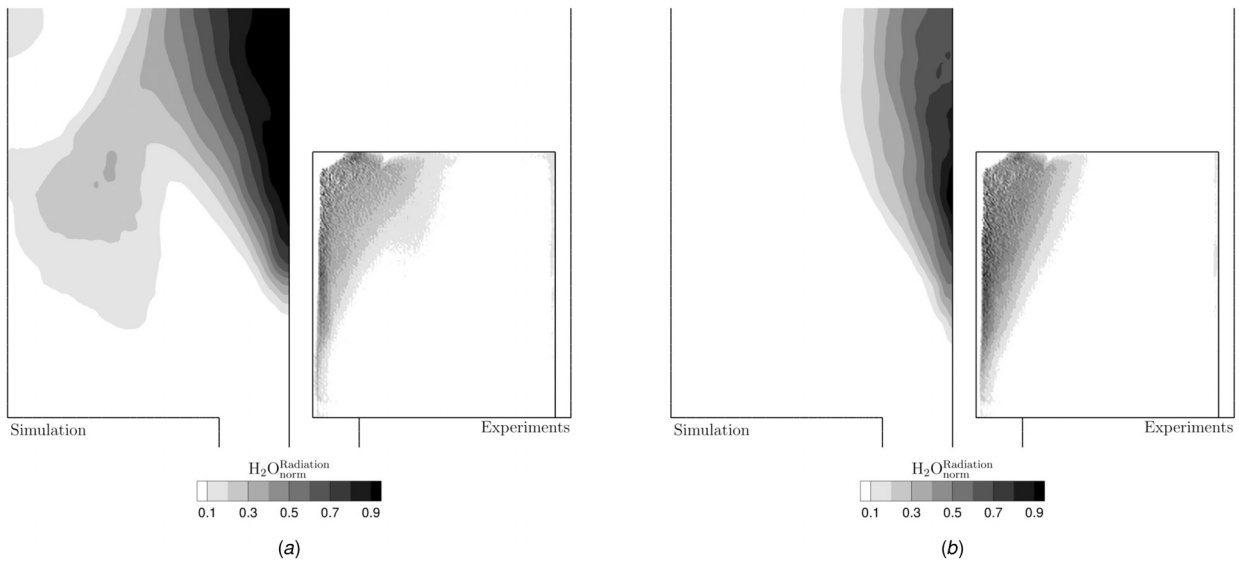

Fig. 6 Comparison of mean normalized water vapor radiation intensity from simulations and experiments: (*a*) flame A and (*b*) flame B

thereby enables some mixing of hydrogen and air prior to combustion. Mixing between fuel and air is further enhanced by a large difference in axial and radial (not shown) velocities and density between fuel and air streams. The mean heat release for the globally leaner flame B is almost entirely within the lean branch. For flame A, the locally rich region extends significantly further downstream and covers a substantial part of the heat release region. The reduction of the lateral flame front for flame B can be associated with the lower equivalence ratio in this region. While it is close to stoichiometry for flame A, it is close to the lean flammability limit for flame B.

Table 3 Pressure loss Δp (%)

	Experiments	Simulation
Flame A	2.991	3.467
Flame B	2.932	3.398

Despite the higher global equivalence ratio, flame A anchors slightly further downstream than flame B. This appears to be related to the different relative velocity between the fuel and the inner air stream. This can be further analyzed using the axial velocity presented in Fig. 7(*b*). Due to the same air mass flow, the overall flow field is similar in both configurations. However, the lower hydrogen mass flow rate for flame B leads to a lower fuel velocity and consequently a larger velocity difference between fuel and air stream. This difference enhances the turbulent transport of the hydrogen into the central swirling air jet. Consequently, the lean mixture fraction limit in Fig. 7(*a*) is located further upstream for flame B. This is also confirmed by the hydrogen mass fraction distribution presented in Fig. 7(*d*).

The flow slightly opens up at the combustion chamber entrance due to the induced swirl. Thereby, a narrow central recirculation

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zone is formed in the diffusor. A less pronounced outer recirculation zone is observed in the combustion chamber. While the axial velocity is similar in the air swirlers, significantly larger values occur in the fuel supply of flame A due to the larger hydrogen mass flow rate. Further differences can be identified for the central recirculation zone. While the axial extension of the central recirculation zone is very similar for both conditions, the radial spread is slightly larger for flame B, combined with higher negative values. A larger spread, and consequently enhanced central recirculation, might be suppressed for flame A by the higher axial momentum of the fuel stream.

The mean temperature is presented in Fig. 7(*c*). The highest temperatures are observed on the burner axis downstream of the central recirculation zone. The temperatures are overall lower for flame B due to the leaner equivalence ratio and consequently lower thermal power. The temperature increase on the burner axis is located further upstream for flame B due to the different axial flame location as described above. At the upstream corner of the

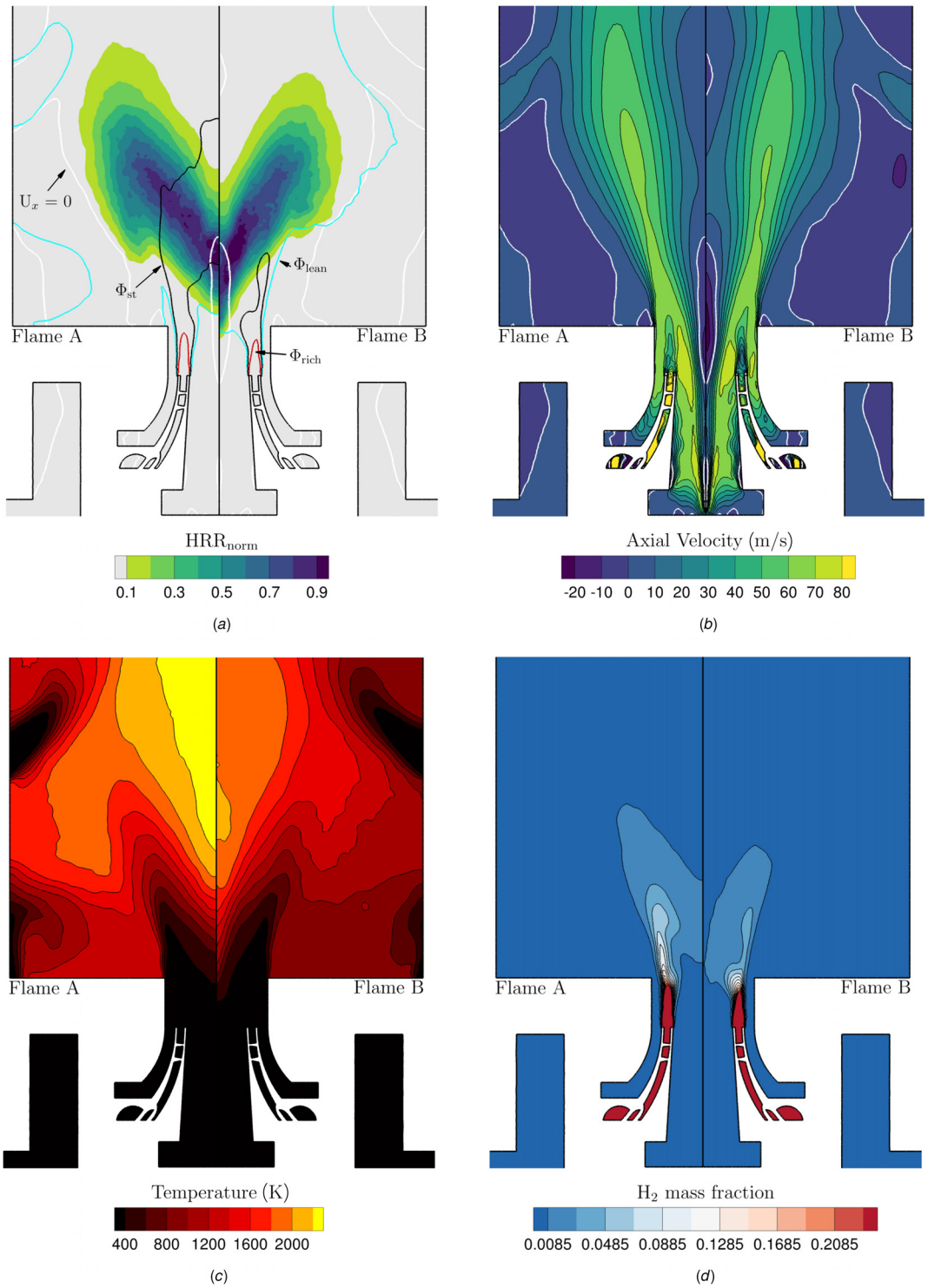

Fig. 7 Comparison of mean fields for flame A (left) and flame B (right): (*a*) normalized heat release rate with isocontours of zero axial velocity and lean, rich, stoichiometric mixture fraction, (*b*) axial velocity with isocontour of zero axial velocity, (*c*) temperature, and (*d*) H_2 mass fraction

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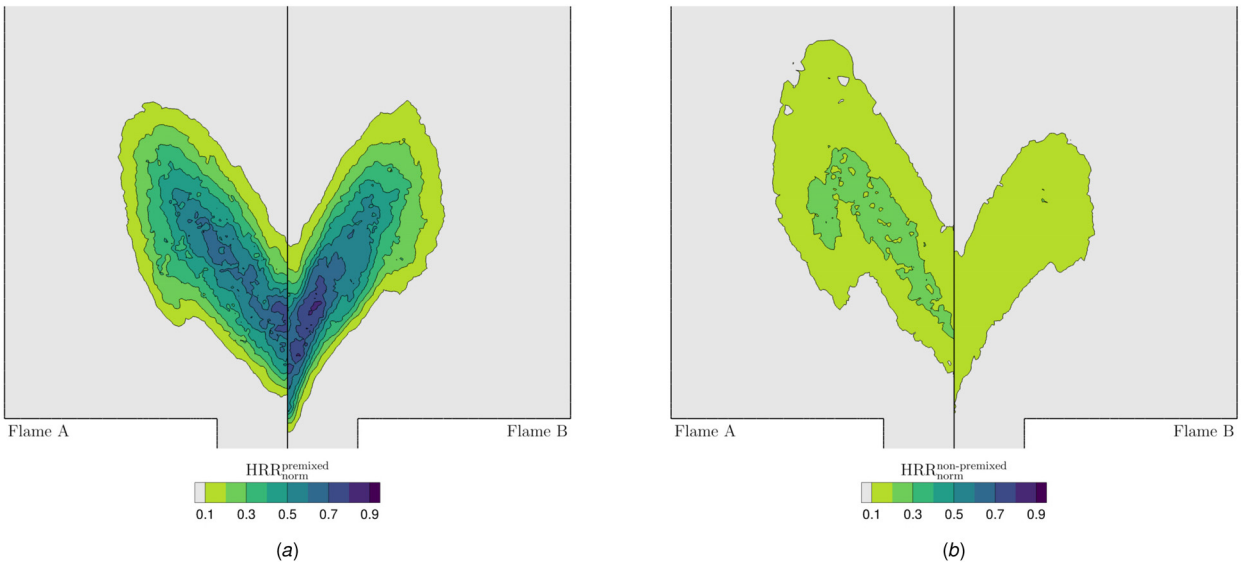

Fig. 8 Comparison of normalized mean heat release rate for flame A (left) and B (right): (*a*) premixed regime and (*b*) nonpremixed regime

combustion chamber and the downstream dilution ports, the reduced temperature due to the cooling air injection is clearly visible.

Although not strictly transferable due to the different flame stabilization, the velocity fields predicted by the simulation can be used to further analyze the flame stabilization of the experiments. When the OH* radical chemiluminescence results are mapped onto the velocity fields from the simulation, the flame stabilizes in the shear layer between fuel and air at an axial location where the velocity difference between the two flows is sufficiently reduced to match the turbulent flame speeds reported in Ref. [49]. This again indicates the shear layer flame stabilization of a turbulent swirling jet diffusion flame.

4.3 Flame Regime. The flame regime is further analyzed based on the simulation results. The simulations in this paper have been performed using a combustion model for partially premixed combustion. However, hydrogen and air are injected separately in the current configuration. Even though the flame lift off allows for a certain mixing between fuel and air prior to combustion, multi-regime combustion characteristics are expected.

The following analysis is based on the Takeno flame index [22], which indicates premixed or nonpremixed regimes based on the gradients of fuel and air. As the flame regime is only relevant in the flame region, the following approach is adopted: The HRR is conditioned by the local flame index. It is thereby split into a contribution from premixed and nonpremixed combustion regime. The time-averaged conditioned HRR fields are presented in Fig. 8 along with the total HRR in Fig. 7(*a*). All fields are normalized with the maximum total HRR. A significant contribution of the total HRR in the primary zone of the configuration is associated with the premixed combustion regime. On the other hand, the HRR associated with the nonpremixed regime is about an order of magnitude lower. Furthermore, the nonpremixed regime tends to occur slightly further downstream. While the general occurrence is similar for the two operating conditions, the contribution of the nonpremixed regime is further reduced for flame B. This can be related to the leaner operating conditions. This indicates an increasing importance of the nonpremixed regime with a further increase in fuel mass flow and, in particular, rich operation.

As discussed above, the comparison of the OH* distribution in Fig. 4 indicates that the local flame stabilization is not accurately captured by the simulation. The different radial position of the flame stabilization may indicate difficulties of the modeling approach to correctly predict the local combustion regime. Since the flame index depends directly on the gradients predicted by the combustion model, the flame regime prediction should be interpreted carefully.

5 Conclusion

The combustion of pure hydrogen in a dual-swirl burner for aero-engine applications has been investigated by experiments and LES. Two lean operating conditions with different equivalence ratios and thermal power but constant air mass flow are considered.

First, the capability of the simulation to predict the flame position and shape is investigated by comparison of OH* radical chemiluminescence measurements and the LES fields. Two distinct reaction fronts are observed for flame A: an upstream inner front and a lateral flame front further downstream, which are both predicted by the simulation. For flame B, the intensity of the lateral flame front is significantly reduced but is slightly more pronounced in the LES results. Despite the leaner equivalence ratio, flame B stabilizes further upstream in both simulation and experiment. This can be explained by the different relative velocities between fuel and air stream which enhance the turbulent transport of fuel into the central air stream. Due to the low density of hydrogen, changes in the fuel mass flow result in significant aerodynamic effects.

While the lift of heights of both flames and the global flame spread are reasonably predicted by the simulations, the flame stabilization mechanism appears to be different between simulation and experiment. The combustion model used in this work is designed for the partially premixed combustion regime and has been successfully used in previous work for technically premixed configurations. Since fuel and air are injected separately in the current configuration, multiregime combustion is expected to play a more important role. In the experiments, the flame stabilizes in the shear layer between the fuel and air stream, resembling the flame stabilization mechanism of a turbulent swirling jet diffusion flame. The combustion model does not capture this flame stabilization mechanism and instead predicts the flame front on the burner centerline.

The analysis of the premixed and nonpremixed regime based on the flame index conditioned heat release rate indicates that the importance of the nonpremixed regime tends to increase with increased fuel mass flow. The ability of the combustion model to correctly predict the nonpremixed flame stabilization mechanism needs to be improved in order to use CFD as a reliable design tool and predict the correct flame behavior for a wide range of operating conditions. In addition, as preferential diffusion effects become more important in hydrogen flames, a model extension to account for

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nonunity Lewis number effects could further improve the predictive capabilities of the simulation.

Acknowledgment

We acknowledge the funding by the DLR's aeronautics directorate through the project ADAPT ("Assessment and Digitalization of Forthcoming Propulsion Technologies") and the German Federal Ministry for Economic Affairs and Climate Action (BMWK) through the project WotAn (FKZ 20M2104B).

Funding Data

- DLR's aeronautics directorate through the project ADAPT ("Assessment and Digitalization of Forthcoming Propulsion Technologies") (Funder ID: 10.13039/501100002946).
- German Federal Ministry for Economic Affairs and Climate Action (BMWK) through the project WotAn (FKZ 20M2104B; Funder ID: 10.13039/501100006360).

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

Roman Letters

$\dot{m}$ = mass flow (g/s)
P_{th} = thermal power (kW)
T = temperature (K)

Greek Symbols

Δp = relative pressure loss (%)
Φ_{PZ} = primary zone equivalence ratio

Abbreviations

CFD = computational fluid dynamics
DTF = dynamic thickened flame
FGM = flamelet-generated manifolds
HRR = heat release rate
IR = infrared
LES = large-eddy-simulations
UV = ultraviolet

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